

VANNES SENE, France

WMO: 072100

Lat: 47.6045N

Lon: 2.7140W

Elev: 3

StdP: 101.29

Time Zone: 1.00 (EUC)

Period: 99-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-2.6	-1.1	-5.8	2.3	-0.1	-4.0	2.7	0.7	11.6	9.1	10.8	8.2	2.3	40	0.543

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.9	28.9	20.2	26.5	19.1	24.7	18.3	20.8	26.7	19.9	25.2	19.0	23.1	3.8	260

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
19.0	13.8	22.5	18.1	13.0	21.5	17.3	12.4	20.6	60.0	26.7	56.7	25.3	54.0	23.1	26.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 9.8	8.5	7.6	-4.5	32.1	1.6	2.6	-5.7	34.0	-6.6	35.5	-7.5	37.0	-8.7	38.9		
(5)			-5.2	22.7	2.0	1.4	-6.6	23.7	-7.8	24.5	-9.0	25.3	-10.4	26.4		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	12.0	6.4	6.6	8.4	10.5	13.4	16.6	18.4	17.6	16.2	12.9	9.0	7.1		
	DBStd	5.25	3.27	3.08	2.97	2.95	2.89	3.22	2.83	2.67	2.89	2.94	2.93	3.32		
	HDD10.0	479	117	99	67	28	4	1	0	0	1	9	53	100		
	HDD18.3	2418	371	328	309	236	155	69	32	45	76	168	280	349		
	CDD10.0	1193	5	4	16	42	111	199	261	234	187	100	23	9		
	CDD18.3	89	0	0	0	0	3	18	36	21	12	1	0	0		
	CDH23.3	724	0	0	0	3	25	157	308	149	82	0	0	0		
	CDH26.7	184	0	0	0	0	2	42	88	36	17	0	0	0		
(14) Wind	WSAvg	3.8	3.8	4.1	4.2	4.0	3.8	3.6	3.7	3.4	3.4	3.5	3.6	3.9		
(15) Precipitation	PrecAvg	895	110	77	70	70	61	45	54	49	55	106	107	104		
	PrecMax	1122	210	186	198	154	108	102	91	120	175	164	235	210		
	PrecMin	592	22	18	8	21	20	12	9	10	14	17	28	26		
	PrecStd	124	57	43	44	36	24	27	22	31	38	43	53	58		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	13.4	15.1	18.7	23.3	26.1	31.6	32.8	30.2	29.1	21.8	17.0	14.3		
		MCWB	12.0	11.0	13.3	16.0	18.6	22.5	20.5	20.4	20.0	17.0	14.8	12.9		
	2%	DB	12.7	12.9	16.0	20.3	23.2	27.1	29.0	27.0	25.6	19.8	15.7	13.4		
		MCWB	11.5	10.6	11.7	14.2	17.3	20.0	19.8	19.2	18.3	16.3	14.1	12.3		
	5%	DB	12.0	12.0	14.1	17.7	21.0	24.8	26.8	24.8	23.2	18.3	14.9	12.9		
		MCWB	10.9	10.3	10.7	12.8	15.9	18.5	19.2	18.2	17.4	15.6	13.4	11.9		
	10%	DB	11.2	11.2	12.8	15.6	18.9	22.8	24.4	22.8	21.5	17.3	13.9	12.1		
		MCWB	10.2	9.9	10.5	12.0	14.7	17.3	18.4	17.5	16.7	15.2	12.5	11.1		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	12.6	12.2	13.8	16.8	19.5	23.3	21.9	21.6	20.7	18.0	15.9	13.5		
		MCDB	13.0	13.6	16.9	22.0	24.9	29.4	28.8	27.6	26.4	20.2	16.4	13.8		
	2%	WB	11.9	11.4	12.5	14.8	17.8	20.5	20.7	20.3	19.4	17.0	14.7	12.8		
		MCDB	12.5	12.4	14.5	18.7	22.2	26.0	26.9	25.4	23.7	18.6	15.3	13.3		
	5%	WB	11.2	10.9	11.7	13.5	16.4	19.2	19.7	19.3	18.4	16.4	13.8	12.2		
		MCDB	11.9	11.6	13.3	16.7	20.1	23.6	25.6	23.1	21.9	17.8	14.6	12.7		
	10%	WB	10.5	10.2	11.0	12.4	15.2	18.0	18.8	18.3	17.6	15.7	12.7	11.4		
		MCDB	11.2	11.0	12.4	15.1	18.1	21.9	23.1	21.3	20.6	17.1	13.8	12.1		
(35) Mean Daily Temperature Range	5% DB	MDBR	5.9	6.9	7.6	8.8	9.0	9.9	9.9	9.9	9.7	7.8	7.2	6.1		
		MCDBR	5.4	7.3	9.6	12.2	12.2	13.2	14.2	12.6	11.9	8.2	6.9	5.3		
	5% WB	MCWBR	4.8	5.5	6.2	7.0	7.0	6.8	6.4	6.0	5.8	5.4	5.5	4.8		
		MCWBR	5.1	6.1	7.7	10.4	11.2	11.9	12.3	10.7	10.1	6.9	6.0	5.0		
(40) Clear-Sky Solar Irradiance	taub		0.336	0.350	0.378	0.393	0.398	0.397	0.393	0.382	0.367	0.365	0.342	0.329		
		taud	2.411	2.384	2.312	2.285	2.317	2.351	2.366	2.414	2.437	2.433	2.426	2.392		
	Ebn at Noon		734	806	835	858	866	867	865	861	841	778	719	690		
		Edh at Noon	70	87	109	123	123	120	117	107	95	82	68	63		
(44) All-Sky Solar Radiation	RadAvg		1.06	1.99	3.19	4.77	5.65	6.21	5.93	5.17	4.11	2.34	1.34	0.92		
	RadStd		0.10	0.19	0.32	0.47	0.38	0.56	0.26	0.34	0.28	0.19	0.11	0.10		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (29 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page